

Dear Editor,

We are pleased to submit our manuscript entitled “Competing feedback channels drive phase transitions in collective emotion” for consideration in *Chaos, Solitons & Fractals*.

Social systems exhibit collective transitions—prolonged quiescence followed by abrupt polarization—that bear the hallmarks of nonlinear dynamics yet lack a mechanistic account of *when* instability arises endogenously. We address this by introducing a minimal mixed-feedback model in which a stabilizing (mainstream) channel competes with an amplifying (user-generated) channel. A single control parameter—the channel-balance ratio r —tunes the system across a stability boundary. Mean-field analysis yields an analytic critical point, and a Ginzburg–Landau reduction near criticality produces a quartic effective potential that transitions from a single to a double well, placing the bifurcation in the standard mean-field universality class.

The model predicts two signatures beyond the static bifurcation: (i) direction–intensity entanglement, where elevated activity signals moderate-buffer depletion and heightened fragility before a polarized direction is selected; and (ii) critical slowing down, with autocorrelation rising as the system approaches the stability boundary. Agent-based simulations on Erdős–Rényi and Barabási–Albert networks confirm that these features persist under finite-size fluctuations and topological heterogeneity, with Binder cumulant crossings providing finite-size diagnostics. In large-scale social-media data spanning the COVID-19 period, both predicted signatures appear: segments with elevated activity exhibit larger polarization jumps, and greater user-generated dominance covaries with higher volatility.

We believe this work fits squarely within the scope of *Chaos, Solitons & Fractals*. It contributes a new nonlinear mechanism—feedback-channel competition—to the theory of collective transitions in complex systems, demonstrates the full dynamical toolkit (bifurcation analysis, Ginzburg–Landau reduction, finite-size scaling, critical slowing down, agent-based simulation), and grounds the predictions in empirical data. The direction–intensity decomposition and the channel-balance control parameter offer both analytical insight and practical levers for managing collective resilience.

We confirm that this manuscript has not been published elsewhere and is not under consideration by another journal. All authors have approved the submission.

Thank you for considering our work.

Sincerely,

On behalf of all authors,

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